Smart Medication Dispenser and Alert System

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Embedded Systems

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Introduction

Medication non-adherence is a widespread problem that affects patient health outcomes, particularly among elderly individuals and patients with chronic diseases. Forgetting to take medication, taking incorrect doses, or taking medication at the wrong time can lead to serious medical complications and increased healthcare costs.

The Smart Medication Dispenser and Alert System is designed to address these issues by automating medication dispensing and providing timely alerts to users. The system ensures that medication is dispensed only at scheduled times, accompanied by visual and audible alerts, and requires user confirmation to verify that the medication has been taken.

Main Components:

- **Microcontroller (Arduino UNO R4):**
  I used the Arduino UNO R4 as the main controller of the system. It is the brain of the smart medication dispenser. It controls all other components, reads the time from the RTC module, activates the servo motor to dispense medication, turns on the buzzer and LEDs for alerts, and reads the confirmation button from the user.

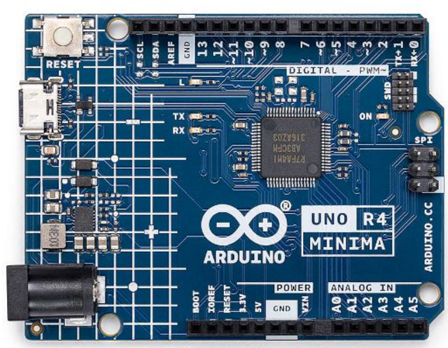

Figure 1: Arduino UNO R4

- **Real-Time Clock (RTC) Module:**
  I used an RTC module to keep track of the current time and date. This component is very important because the system depends on accurate timing to dispense medication at the correct scheduled time. The RTC continues to keep time even if the system is powered off.

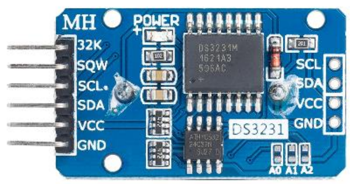

Figure 2: RTC

- **Servo Motor (180°):**
  I used a **180-degree servo motor** to control the medication dispensing mechanism. The servo motor rotates to a specific angle to release the pill from its compartment, then returns to its original position. The Arduino controls the servo using PWM signals.

Figure 3; Servo Motor

- **LCD Display (16×2):**
  I used a **16×2 LCD display** to show useful information to the user. The display shows the

current time, medication reminders, and alert messages such as “Take Your Medication”. This helps the user clearly understand the system status.

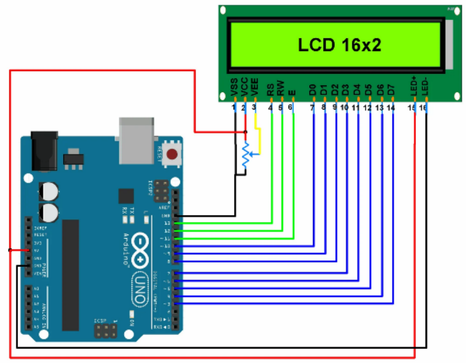

Figure 4: LCD

- **Buzzer:**
  I will use a buzzer to provide an audible alert when it is time to take medication. The buzzer helps notify the user even if they are not looking at the device. It will continue to beep until the user confirms that the medication has been taken.

Figure 5: Active Buzzer

- **LED Light:**
  I used a red LED light as visual indicators. The red LED turns ON when it is time to take medication.

Figure 6: LED

- **Push Button:**
  I used a push button to allow the user to confirm that they have taken the medication. When the button is pressed, the system stops the alert, turns off the red LED.

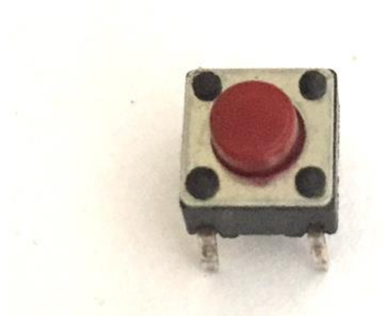

Figure 7: Push- Button

- **Wires:**
  I will use connecting wires to link all components together, including the Arduino, servo motor, LCD, buzzer, LEDs, RTC module, and push button. These wires allow signals and power to flow between components.

Figure 8: Wires

- **Breadboard:**
  I will use a breadboard to connect and test all components without soldering. The breadboard makes it easy to modify the circuit during testing and debugging.

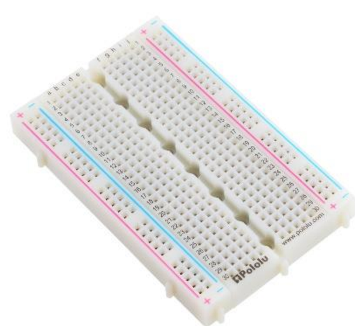

Figure 9: Breadboard

- **Pins in Arduino:**

  - A pin is a small connection point on the Arduino board.

  - We use pins to connect components like sensors, LEDs, motors, and displays.

  - Each pin has a specific function: it can either send or receive electrical signals.

- There are different types of pins: digital, analog, PWM, and power pins.

Types of Pins in Arduino:

Pin Type	What it does
Digital Pins	Used for sending or receiving digital signals, like controlling an LED
Analog Pins	Used to read analog values (like light or temperature sensors)
PWM Pins	These are special digital pins that can send signals like analog (used for servo motors)
Power & GND Pins	Provide power or ground (GND) to other components

Table 1: Arduino Pin Types

- I connected the Servo Motor to a PWM Pin so I can control the the dispenser movement
- The LCD Display was connected to multiple Digital Pins to show the state of the system (Alert, meds taken)

Project Features

- **Automated Medication Dispensing:**
  The system automatically dispenses medication at predefined times without requiring manual intervention, reducing the risk of missed or incorrect doses.
- **Real-Time Time Management:**
  The system uses a real-time clock (RTC) to continuously monitor the current time and ensure medication is dispensed according to an accurate schedule.
- **Servo-Based Dispensing Mechanism:**
  A 180-degree servo motor is used to rotate a medication carousel with four compartments, each separated by 45 degrees, allowing controlled and precise dispensing.
- **Multi-State System Operation:**
  The system operates using four logical states: Idle, Dispense, Alert, and Confirmed, ensuring organized and reliable system behavior.
- **Visual Alert System:**
  A red LED provides a clear visual indication when medication is ready to be taken, making the system accessible to users with hearing difficulties.

- **Audible Alert System:**
  A buzzer generates sound alerts to notify the user when it is time to take medication, ensuring attention even when the user is not near the device.
- **User Confirmation Mechanism:**
  A push button allows the user to confirm that the medication has been taken, improving accountability and preventing false dose registration.
- **Dose Tracking Capability:**
  The system records when a dose is confirmed as taken, allowing future expansion for logging and adherence monitoring.
- **Minute-Based Scheduling:**
  The system checks the time every minute, ensuring reliable triggering of dispensing events without relying on exact second matching.
- **User-Friendly Interface:**
  The system provides simple interaction through LEDs, buzzer, and button, making it suitable for elderly and non-technical users.
- **Low-Cost and Scalable Design:**
  The project uses affordable and widely available components, making it suitable for home use and scalable for small healthcare facilities.
- **Reliable Embedded Control:**
  The system is controlled by the Arduino UNO R4 Minima, ensuring stable performance and ease of future modification or expansion.

Flowchart of the system operation

The following flowchart illustrates the main operational flowchart of the smart medication dispenser system. The system operates in four states: Idle, Dispense, Alert, and Confirmed. It continuously checks the time using the RTC, dispenses medication at scheduled times, alerts the user, and returns to the idle state after confirmation.

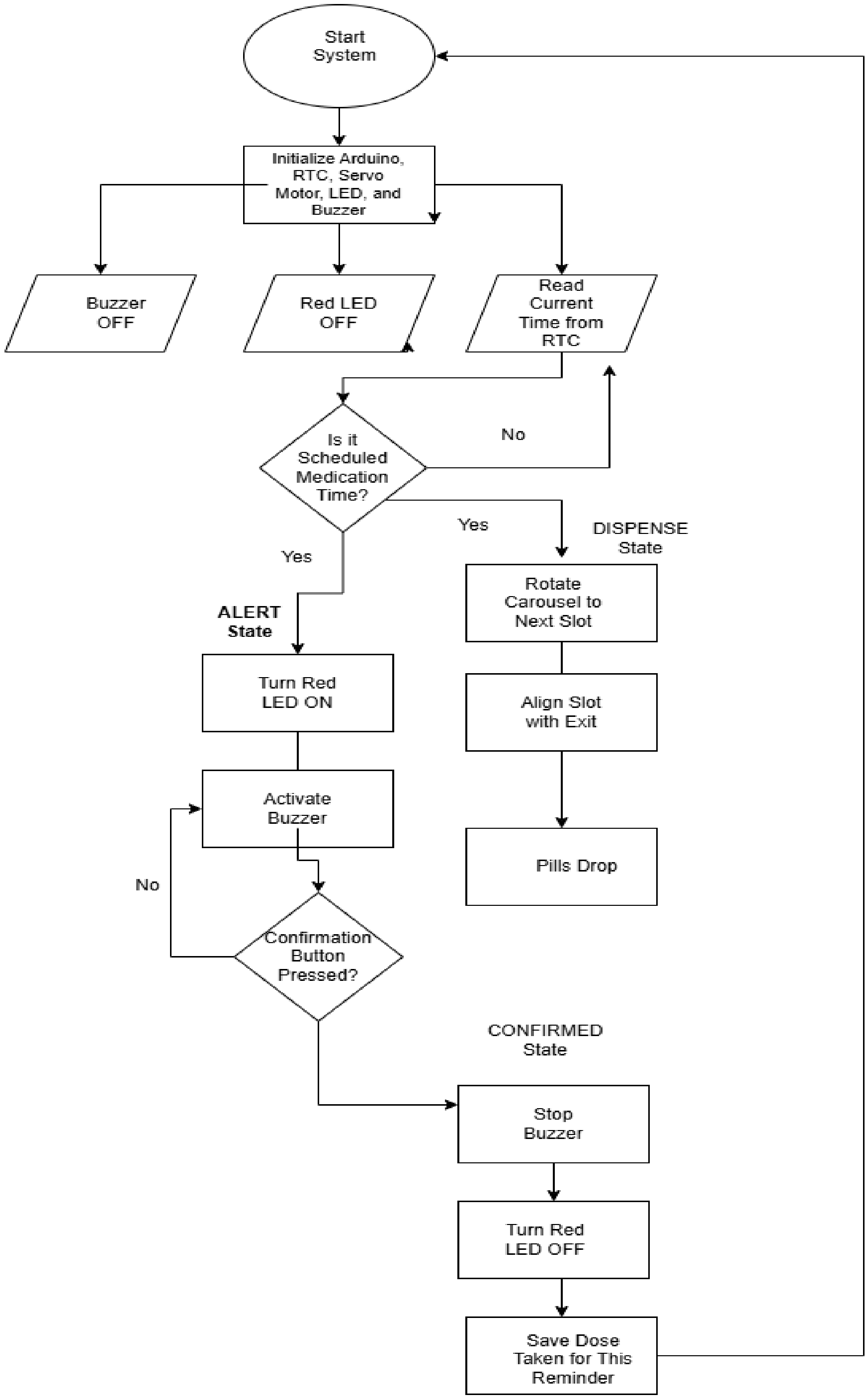

Table 2: Flowchart

1. IDLE (waiting)

- RTC keeps checking time
- Red LED OFF, buzzer OFF

2. DISPENSE (at scheduled time)

- Rotate carousel to align the next slot with the exit
- Pills drop

3. ALERT (user needs to take pill)

- Red LED ON (or blinking)
- Buzzer beeps (or ON)
- System waits for confirmation button

4. CONFIRMED (pill taken)

- Stop buzzer
- Turn Red LED OFF
- Save “dose taken” for this reminder
- Return to **IDLE** and wait for next reminder time

Description of Actuators and Transducers Used in the System

This section describes the main electronic components used in the Smart Medication Dispenser and Alert System, focusing on each component’s role, why it was selected, and how it interacts with the rest of the system. The design emphasizes reliability, ease of use, and low-cost implementation for home healthcare environments.

Microcontroller Unit: Arduino UNO R4 (Main Controller)

The system is controlled by the **Arduino UNO R4** board, which functions as the “brain” of the device. It executes the program logic, reads time information from the real-time clock (RTC), controls the dispensing actuator (servo motor), and manages the user interface components (LED, buzzer, and button).

Key roles in the system:

- Continuously reads the current time from the RTC.
- Compares current time with the reminder schedule (minute-based triggering).
- Sends control signals (PWM) to the servo motor to rotate to dispensing angles.
- Activates the red LED and buzzer during alerting.
- Reads the confirmation push button to stop alerts and confirm dose intake.

Why Arduino UNO R4 was selected:

- Easy programming using Arduino IDE.
- Reliable digital I/O for button, buzzer, and LED control.
- PWM support for servo motor control.
- Suitable for embedded real-time control tasks.

Real-Time Clock (RTC): Timekeeping and Scheduling Component

The RTC is responsible for maintaining accurate time so the system can dispense medication at the correct schedule. The RTC enables consistent reminder operation even when the microcontroller is reset (depending on RTC configuration and backup).

Key roles in the system:

- Provides accurate time (hours/minutes/seconds).
- Enables scheduled dispensing without relying on the Arduino's internal timing alone.
- Supports stable time monitoring for long-term operation.

Importance for medication systems:
Medication adherence requires consistent timing. Using an RTC reduces timing drift and improves schedule reliability compared to using only software delays or internal counters.

Dispensing Actuator: 180° Servo Motor (Carousel Rotation)

A **180-degree servo motor** is used as the mechanical actuator that rotates the medication carousel/dispenser mechanism. Unlike stepper motors, a servo motor provides **simple and accurate angle control** using a PWM signal, making it suitable for a compact dispenser.

How it is used in this project:

- The servo rotates to predefined angles to align a compartment with the dispensing exit.
- The system uses multiple compartments; for a 4-slot design, angles are typically set such as **0°, 45°, 90°, 135°** (based on the physical layout).
- After aligning and allowing pills to drop, the servo returns to a home position.

Advantages of using a servo motor:

- Precise positioning without needing a separate motor driver board.
- Simple wiring (Power, Ground, Signal).
- Controlled movement reduces mechanical errors.

- Lower implementation complexity for student projects.

Output Actuators: LED and Buzzer (User Alerts)

The system uses **visual and audible actuators** to ensure the user receives clear reminders.

Red LED (Visual Alert)

The red LED is a visual indicator that turns ON (or blinks) when a dose is ready.

Role:

- Provides a clear "take medication now" indicator.
- Supports users who may not hear the buzzer.

Buzzer (Audible Alert)

The buzzer provides sound alerts when dispensing occurs and continues until the user confirms.

Role:

- Ensures the user notices the reminder even when not looking at the device.
- Can be configured for continuous tone or periodic beeps to improve attention without being excessively loud.

Input Transducer: Push Button (User Confirmation)

A push button is used to confirm medication intake and stop the alert. This makes the system interactive and supports basic adherence tracking.

Role in the system:

- When pressed, the Arduino stops the buzzer and turns OFF the red LED.
- The system records the dose as "taken" for that reminder cycle.
- The system returns to the IDLE state and waits for the next scheduled dispensing time.

Why confirmation matters:
Without confirmation, the system cannot distinguish between "dispensed" and "taken." User confirmation adds safety and improves reliability for real-world use.

Supporting Prototyping Components: Breadboard and Jumper Wires

Breadboard

Used for building and testing circuits without soldering, allowing rapid modification during development.

Jumper Wires

Provide electrical connections between Arduino pins and components (LED, buzzer, button, RTC, servo), enabling signal and power distribution throughout the system.

Conclusion and Economic Impact

The Smart Medication Dispenser and Alert System presented in this project demonstrates how a simple embedded system can significantly improve medication adherence and patient safety. By integrating an Arduino UNO R4, a 180-degree servo motor, a real-time clock, and basic input/output components, the system successfully automates medication dispensing, alerts the user at scheduled times, and requires confirmation to ensure that medication is taken as intended.

From a functional perspective, the system reduces reliance on human memory and manual supervision, which are common causes of missed or incorrect medication doses. The use of a state-based control approach (Idle, Dispense, Alert, and Confirmed) ensures organized and reliable system behavior, making the solution suitable for daily use by elderly individuals and patients with chronic conditions.

Economic Impact

In an economic context, the Smart Medication Dispenser and Alert System offers several important benefits. Medication non-adherence is a major contributor to increased healthcare costs, including unnecessary hospital admissions, emergency visits, and prolonged treatments. By ensuring that medication is taken on time and in the correct order, this system has the potential to reduce such costs and improve overall healthcare efficiency.

The system is designed using low-cost and widely available components, such as the Arduino UNO R4, a servo motor, LEDs, and a buzzer. This makes the device affordable for home use and accessible to individuals and families without requiring expensive commercial medical equipment. Compared to advanced smart dispensers available on the market, this solution provides essential functionality at a significantly lower cost.

Furthermore, the system can reduce the need for continuous caregiver supervision, which may lower labor costs in home-care and assisted-living environments. Its modular design also allows for future expansion, such as data logging or remote monitoring, without major additional investment.

In conclusion, the Smart Medication Dispenser and Alert System represents a cost-effective, reliable, and scalable solution that addresses both healthcare and economic challenges. By combining automation, user interaction, and accurate time management, the system contributes to improved patient outcomes while helping reduce long-term healthcare expenditures.